

Jacob Darwin I

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ü Medium Portfolio

Professional Summary

Artificial Intelligence and Data Science undergraduate passionate about Generative AI, LLM applications, and intelligent automation systems. Skilled in Python, Retrieval-Augmented Generation (RAG), vector databases, and AI agent workflows. Familiar with OpenAI, Groq, and Claude APIs for building AI-powered applications and exploring prompt engineering, LangChain, and LangGraph concepts.

Technical Skills

Programming: Python,
Generative AI: LLMs, Prompt Engineering, RAG, AI Agents, MCP, OpenAI API, Groq API, Claude API
Frameworks & Tools: LangChain, LangGraph, LlamaIndex
Machine Learning: Machine Learning, NLP Fundamentals
Databases: Vector Databases

Projects

RAG-Based PDF Chatbot | Python, OpenAI API, LangChain, Vector DB

- Built a Retrieval-Augmented Generation chatbot capable of answering questions from uploaded PDF documents.
- Implemented semantic retrieval workflows using vector embeddings and prompt engineering techniques.
- Integrated Groq APIs with LangChain pipelines for conversational question-answering systems.

Deepfake Image & Video Detection using Vision Transformers | Python, Vision Transformer (ViT)

- Developed an AI-powered deepfake detection system capable of identifying manipulated images and videos using Vision Transformer (ViT) architectures.
- Fine-tuned transformer-based deep learning models on real and synthetic media datasets to improve detection accuracy and generalization performance.
- Implemented preprocessing pipelines for frame extraction, image augmentation, and feature embedding generation for video analysis workflows.

AI Agent Workflow Assistant | LangGraph, Python, MCP

- Explored AI agent workflows using LangGraph and Model Context Protocol (MCP) concepts.
- Designed multi-step agent interactions for task automation and intelligent response generation.
- Experimented with tool-calling workflows and prompt orchestration techniques.

Education

Jeppiaar Engineering College
B.Tech in Artificial Intelligence & Data Science

2022 – 2026
CGPA 7.8

Certifications

- Machine Learning A-Z – Udemy
- Microsoft – Introduction to MS Excel

Technical Writing

- Published technical articles on Medium related to Artificial Intelligence and emerging technologies.